FUNDAMENTALS OF DATA ANALYTICS LAB

WEEK 8 ASSIGNMENT

Overview: Company Y is a growing analytics-driven organization that collects data related to customer purchases, employee performance, and marketing campaigns. Management wants to use statistical techniques to make informed decisions instead of relying only on raw observations.

Objective: As a data analyst, apply statistical foundations using Python to analyze patterns, test hypotheses, and generate insights.

Dataset Description

The following five datasets are provided for analysis:

Dataset	File Name	Columns
Sales Data	*sales_data.csv*	Customer_ID, City, Order_Amount, Payment_Method
Employee Performance	*employee_performance.csv*	Employee_ID, Department, Before_Training_Score, After_Training_Score
Survey Data	*survey_data.csv*	Gender, Preferred_Product_Category
Regional Sales	*regional_sales.csv*	Region, Sales
Student Behavior	*student_behavior.csv*	Study_Hours, Marks

Part 1: Descriptive Statistics

Tasks:

- **Calculate the following statistical measures:**
  - Mean
  - Median
  - Variance
  - Standard Deviation
- Compare mean vs. median to identify skewness in the data.
- Identify which dataset has the highest variability.
- Interpret the business meaning of dispersion.

Part 2: Hypothesis Testing (t-test)

> *Context: The company introduced a training program and wants to evaluate its effectiveness.*

Tasks:

- **Perform a paired t-test on the following columns from employee_performance.csv:**
  - Before_Training_Score
  - After_Training_Score
- State the hypotheses:
  - Null Hypothesis (H)
  - Alternative Hypothesis (H)
- Calculate the following:
  - t-statistic
  - p-value
- Interpret results at the 95% confidence level ($\alpha = 0.05$).
- Decide whether the training program is effective based on the results.

Part 3: Chi-Square Test

> *Context: The company wants to determine whether gender influences product preference.*

Tasks:

- Create a contingency table from survey_data.csv.
- Perform the chi-square test of independence.
- Compute the following:
  - Expected frequencies
  - Chi-square statistic
- Interpret whether the two variables are independent.
- Check and validate the assumptions of the chi-square test.

Part 4: ANOVA (Analysis of Variance)

> *Context: The company wants to compare sales performance across different regions.*

Tasks:

- Perform a one-way ANOVA test on regional_sales.csv.
- State the hypotheses:
  - Null Hypothesis (H)
  - Alternative Hypothesis (H)
- Compute the F-statistic and p-value.

- Interpret whether sales differ significantly across regions.
- Suggest actionable next steps if significant differences are found.

Part 5: Correlation Analysis

> *Context: Analyze the relationship between study hours and academic performance using student_behavior.csv.*

Tasks:

- Compute the following correlation coefficients:
  - Pearson correlation
  - Spearman correlation
- Compare both results and discuss differences.
- Identify the strength and direction of the relationship.
- Interpret whether the relationship is linear or monotonic.

Part 6: Advanced Statistical Analysis

Tasks:

- Identify the presence of outliers using appropriate statistical methods.
- Analyze how outliers affect the following:
  - Mean
  - Standard deviation
- Create a scenario where the mean is misleading due to outliers.
- Compare results before and after removing outliers.

Part 7: Statistical Decision Making

Tasks:

Based on your statistical analysis, answer the following business questions and justify each answer with statistical evidence:

- Should the company continue its training programs?
- Which region requires the most improvement?
- Is product preference dependent on gender?

Part 8: Visualization

Required Plots:

- Distribution of Order_Amount — Histogram (from sales_data.csv)
- Regional Sales Comparison — Box Plot (from regional_sales.csv)
- Study Hours vs. Marks — Scatter Plot (from student_behavior.csv)
- Correlation Heatmap — across all relevant numeric variables

Final Task: Statistical Analysis Report

Prepare a comprehensive Statistical Analysis Report that includes the following sections:

#	Section	Description
1	**Descriptive Statistics Summary**	Summary of mean, median, variance, and standard deviation across datasets
2	**Hypothesis Testing Results**	t-test results, chi-square findings, and ANOVA outcomes with interpretations
3	**Correlation Insights**	Pearson and Spearman correlation values with relationship interpretation
4	**Outlier Analysis**	Detection methodology, impact on metrics, and before/after comparison
5	**Final Business Recommendations**	Data-driven recommendations for Company Y based on all analyses